Q - 1) If $y(x) = x$ is a soln. of the D.E

$$y'' - \left(\frac{2}{x^2} + \frac{1}{x}\right)(xy' - y) = 0$$

then find its general soln.

Soln) Given $y_1 = x$

then second soln $y_2 = u y_1$

on comparing D.E with $y'' + P(x)y' + Q(x)y = 0$

$$P(x) = -\left(\frac{2}{x} + 1\right)$$

$$u = \int \frac{1}{y_1^2} e^{-\int P(x)dx}$$ [Using Reduction of order Method]

$$u = \int \frac{1}{x^2} e^{+\int\left(\frac{2}{x}+1\right)dx}$$

$$= \int \frac{1}{x^2} e^{[2\log x + x]} dx$$

$$= \int \frac{1}{\cancel{x^2}}[\cancel{x^2} \cdot e^x] dx = e^x$$

$$y_2 = u y_1 = x e^x$$

General soln $y = c_1 y_1 + c_2 y_2$

$$= c_1 x + c_2 x e^x$$

$$= x[c_1 + c_2 e^x] \quad \text{(Ans)}$$

Q-2 Solve the I.V.P

$$6y'' - y' - y = 0 \qquad y(0) = -0.5$$
$$y'(0) = 1.25$$

Soln Assume $y = e^{rx}$ then

$$(6r^2 - r - 1)e^{rx} = 0$$
$$6r^2 - r - 1 = 0 \rightarrow \text{Characteristic Eq}^n$$
$$r = \frac{1 \pm \sqrt{1+24}}{12} = \frac{1 \pm 5}{12}$$

$$r_1 = \frac{1}{2} \qquad r_2 = -\frac{1}{3}$$

$$y(x) = c_1 e^{\frac{x}{2}} + c_2 e^{-\frac{x}{3}}$$

on applying initial conditions

$$y(0) = c_1 + c_2 = -0.5$$
$$y'(x) = \frac{1}{2} c_1 e^{x/2} - \frac{1}{3} c_2 e^{-x/3}$$

$$y'(0) = \frac{1}{2} c_1 - \frac{1}{3} c_2 = 1.25$$

$$c_1 + c_2 = -0.5$$
$$\frac{1}{2} c_1 - \frac{1}{3} c_2 = 1.25$$
$$c_1 = \frac{13}{10} \qquad c_2 = -\frac{9}{5}$$

~~General~~ Solution

$$y(x) = \frac{13}{10} e^{x/2} - \frac{9}{5} e^{-x/3} \quad \text{(Ans)}$$

Q-3 Deduce whether following functions are linearly dependent or linearly Independent. Show the details of your work

a) $x^2, x^2 \ln x$; $x > 1$

b) $\sin 2x$, $\cos x \cdot \sin x$; $x < 0$

Soln a) Assume x^2 and $x^2 \ln x$ are dependent

then $x^2 \ln x = Cx^2$ $\forall x > 1$

where C is a non zero constant.

$\because x > 1$

divide by x^2 both side

$$\frac{x^2 \ln x}{x^2} = \frac{Cx^2}{x^2}$$

$\Rightarrow \ln x = C$ (const) $\forall x > 1$

which is a contradiction

$\Rightarrow x^2$ and $x^2 \ln x$ are L.I

b) $\sin 2x$ and $\cos x \cdot \sin x$ are L.D

$\because \sin 2x = 2(\sin x \cdot \cos x)$ $\forall x \in \mathbb{R}$

for $x < 0$

$\sin 2x = 2(\sin x \cdot \cos x)$ holds

with $c = 2$. (constant)

Q-4 If $y(x)$ be the solution of I.V.P

$$y'' + 4y' + 4y = 0$$

$y(0) = 0 \quad y'(0) = 2$ then value of

$y(\log 2) = \quad ?$

Soln) Assume $y = e^{rx}$

$(r^2 + 4r + 4) e^{rx} = 0$

$r^2 + 4r + 4 = 0$ Characteristic Eq^n

$r = -2, -2$

General soln

$y = (c_1 + c_2 x) e^{-2x}$

where c_1, c_2 are arbitrary constant

Applying Initial conditions

$y(0) = c_1 = 0$

$y(x) = c_2 x e^{-2x}$

$y'(x) = c_2 [e^{-2x} - 2x e^{-2x}]$

$y'(0) = c_2 [1 - 0] = 2$

$c_2 = 2$

$y(x) = 2x e^{-2x}$ is the soln of D.E

$y(\log 2) = 2 \log 2 \, e^{-2(\log 2)}$

$= \dfrac{2 \log 2}{4} = \dfrac{\log 2}{2}$ (Ans)

_/_/_

Q - 5 Find the soln of I.V.P

$y'' + y' + 2y = 0 \qquad y(0) = 0$

$y'(0) = 2$

Soln) assume $y = e^{rx}$ as the soln of D.E

then

$(r^2 + r + 2) e^{rx} = 0$

$r^2 + r + 2 = 0 \rightarrow$ Characteristic Eqn

$r = \frac{-1 \pm \sqrt{7} i}{2}$

$y(x) = e^{-1/2 x} \left(C_1 \cos \frac{\sqrt{7}}{2} x + C_2 \sin \frac{\sqrt{7}}{2} x \right)$

is the general soln of D.E where C_1 & C_2 are arbitrary constant

applying Initial conditions

$y(0) = C_1 = 0$

$y(x) = C_2 e^{-1/2 x} \sin \frac{\sqrt{7}}{2} x$

$y'(x) = C_2 \Big[e^{-\frac{1}{2}x} \cos \frac{\sqrt{7}}{2} x \cdot \frac{\sqrt{7}}{2}$

$+ \left(\frac{-1}{2} \right) e^{-\frac{1}{2}x} \sin \frac{\sqrt{7}}{2} x \Big]$

$y'(0) = C_2 \left[1 \cdot 1 \cdot \frac{\sqrt{7}}{2} + 0 \right] = 2$

$C_2 = \frac{4}{\sqrt{7}} \qquad y(x) = \frac{4}{\sqrt{7}} e^{-\frac{1}{2}x} \sin \frac{\sqrt{7}}{2} x$ (Ans)

6. Let $y(x)$ be the solution of the following initial value problem:

$$x^2\frac{d^2y}{dx^2} - 4x\frac{dy}{dx} + 6y = 0, \qquad x > 0,$$

$$y(2) = 0, \qquad \frac{dy}{dx}(2) = 4.$$

Then $\quad y(4) =$ _________.

Solution:

We try a Cauchy–Euler type trial solution of the form

$$y = x^m.$$

Then

$$y' = mx^{m-1}, \qquad y'' = m(m-1)x^{m-2}.$$

Substituting these into the ODE

$$x^2y'' - 4xy' + 6y = 0,$$

gives

$$x^2\left(m(m-1)x^{m-2}\right) - 4x\left(mx^{m-1}\right) + 6x^m = 0.$$

Simplifying,

$$m(m-1)x^m - 4mx^m + 6x^m = 0.$$

Factor out $x^m \neq 0$:

$$m(m-1) - 4m + 6 = 0.$$

Thus the indicial equation is

$$m^2 - 5m + 6 = 0,$$

whose roots are

$$m = 3, \qquad m = 2.$$

Hence the general solution is

$$y(x) = c_1x^3 + c_2x^2.$$

Using the initial conditions:

$$\begin{aligned} y(2) &= 8c_1 + 4c_2 = 0, \\ y' &= 3c_1x^2 + 2c_2x, \\ y'(2) &= 12c_1 + 4c_2 = 4. \end{aligned}$$

Solving,

$$c_1 = 1, \qquad c_2 = -2.$$

Thus,

$$y(x) = x^3 - 2x^2,$$

and therefore

$$y(4) = 64 - 32 = 32.$$